Name: _____

TOPIC TEST 2

Chapter 2 Discrete random variables

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Seven multiple-choice questions (7 marks).
- Part B: Eight short-answer questions (44 marks).
- Part C: Three analysis questions (18 marks).

Part A

Multiple-choice questions

Seven multiple-choice questions
1 mark each: 7 marks
Circle the correct answer.

1 A pair of dice is rolled and $X =$ the sum of the numbers. Which of the following ordered pairs would *not* make up the probability function for X?

A $\left(1, \frac{1}{36}\right)$

B $\left(5, \frac{4}{36}\right)$

C $\left(11, \frac{2}{36}\right)$

D $\left(8, \frac{5}{36}\right)$

E $\left(2, \frac{1}{36}\right)$

[1 mark]

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2 Which of the following is *not* an example of a discrete variable?

A The number of people living in a household

B The distance between the bumper bars of cars in a parking lot

C The number of votes polled by a politician in an election

D The number of coins in the pockets of students in a class

E The shoe sizes of students in a class

[1 mark]

3 Which one of the following could *not* represent a probability distribution?

A

x	$p(x)$
7	0.1
8	0.2
9	0.3
10	0.4

B

x	$p(x)$
110	$\frac{3}{8}$
220	$\frac{1}{3}$
330	$\frac{1}{12}$
440	$\frac{1}{4}$

C

x	$p(x)$
5	0.25
10	0.25
15	0.25
20	0.25

D

x	$p(x)$
-8	$\frac{1}{8}$
-5	$\frac{1}{4}$
3	$\frac{3}{8}$
9	$\frac{1}{4}$

D

x	$p(x)$
-8	$\frac{1}{8}$
-5	$\frac{1}{4}$
3	$\frac{3}{8}$
9	$\frac{1}{4}$

[1 mark]

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4 A uniform probability distribution for the discrete random variable X, has the following outcomes:

2 4 5 7 8 9

What is $P(X = 7)$?

A $\frac{1}{7}$

B $\frac{5}{6}$

C $\frac{1}{2}$

D $\frac{1}{6}$

E $\frac{2}{3}$

[1 mark]

5 A discrete random variable, X, can take the values 10, 20, 30 and 40 with probabilities 0.15, 0.2, 0.25 and 0.4 respectively. What is the expected value of X?

A 25

B 32

C 29

D 19

E 27

[1 mark]

6 A coin is weighted so that the chance of a head is $\frac{2}{3}$. The coin is tossed twice. How many tails are expected?

A $\frac{1}{3}$

B $\frac{4}{9}$

C $\frac{2}{3}$

D $\frac{5}{9}$

E $\frac{2}{3}$

[1 mark]

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7 A discrete random variable, X, can take the values 2, 7, 11 and 14 with probabilities 0.2, 0.25, 0.35 and 0.2 respectively. The value of Var(X) is approximately:

A 9

B 13

C 17

D 21

E 6.5

[1 mark]

Part B

Short-answer questions

Eight short-answer questions
44 marks
Show your working where appropriate.

1 The number of applications loaded onto a group of students' smartphones was found to be as follows:

8 6 4 2 9 5 7 8 5 7 5 6 5 6 3 4 5

7 2 1 10 4 8 2 9 5 3 8 5 4 9 7 8 6

If the number of smartphone applications is the variable, X:

a construct a probability distribution for X

b calculate $P(4 \leq x < 6)$

c calculate $E(X)$.

[7 marks]

2 A bag contains six equally sized marbles numbered 1, 4, 4, 6, 7 and 8. Let X be the number of the marble drawn.

a Construct a probability distribution for X.

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b Calculated $E(X)$.

c If the value of the number on each marble is increased by 5, what is the new expected value of X?

d If the value of each of the original numbers on the marbles is increased by a factor of 10, what is the new expected value of X?

[6 marks]

3 A container holds 10 identical computer chips. It is known that three of the chips are defective. A chip is randomly selected from the container, tested and then set aside. Three selections are made in all.

a What is the probability of only one defective chip in three selections?

b What is the probability that the only defective chip will be in the second selection?

[6 marks]

4 Five cards are numbered 1 to 5. A card is selected, not replaced and then another card is selected. Let x be the sum of the numbers selected.

a Construct a probability distribution for X.

b Use the probability distribution to calculate the expected value for X.

[5 marks]

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5 A spinner has equally sized segments with the numbers 11, 12, 13, …, 50.

a Describe the probability distribution for X, the number on which the pointer finishes after each spin.

b Calculate $P(x = 17)$.

c Calculate $P(21 \leq x \leq 32)$.

[5 marks]

6 Calculate $E(X)$, Var(X) and σ for the following distribution:

x	7	9	10	12	15
$P(X = x)$	0.09	0.18	0.33	0.21	0.19

[6 marks]

7 A lottery with 240 000 tickets has the following prizes:

24 000 prizes of $1

15 000 prizes of $2

12 000 prizes of $3

1200 prizes of $10

500 prizes of $20

300 prizes of $30

4 prizes of $500

1 prize of $5000

a Find the expected winnings of a ticket.

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b If each ticket costs \$1, what is the expected value of the game?

[3 marks]

8 An unbiased coin is tossed four times. Let X = the number of heads that occur.

a Construct the probability distribution for X.

b Calculate $E(X)$ and compare it with the number of heads your intuition tells you should occur in four tosses of a coin.

[6 marks]

Part C

Analysis questions

Three analysis questions
18 marks
Show your working where appropriate.

1 A player tosses two unbiased coins. The player wins \$3 if two heads are tossed and \$1 if only one head is tossed. For the game to be fair, how much should the player lose if no heads are tossed?

[7 marks]

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2 Prove that $\text{Var}(kX) = k^2\text{Var}(X)$ where k is a constant.

[7 marks]

3 An office manager is considering purchasing a multifunction device to perform the functions of a photocopier, printer and fax machine. One consideration is how often the machine will need repairs. Based on experience, the annual repair history for the multifunction device is known to be as follows:

Number of repairs (X) in a year	0	1	2	3
$P(X = x)$	0.2	0.3	0.4	0.1

The annual cost of the service agreement for the multifunction device is $2000, with an additional cost of $500 for each repair.

Calculate the expected annual cost of servicing and repairing the multifunction device.

[4 marks]

[Total marks: 69]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 Probability distribution =

x	2	3	4	5	6	7	8	9	10	11	12
$p(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Ordered pairs $= \left(2, \frac{1}{36}\right), \left(3, \frac{2}{36}\right), \left(4, \frac{3}{36}\right), \left(5, \frac{4}{36}\right), \left(6, \frac{5}{36}\right), \left(7, \frac{6}{36}\right), \left(8, \frac{5}{36}\right), \left(9, \frac{4}{36}\right), \left(10, \frac{3}{36}\right), \left(11, \frac{2}{36}\right), \left(12, \frac{1}{36}\right) \left(1, \frac{1}{36}\right)$ is not one of the ordered pairs listed.

$\therefore$ **A**

2 A The number people living in a household can be counted and cannot take on in-between values, so it is discrete.

B The distance between the bumper bars of cars in a parking lot depends on the accuracy of the instrument used to measure distance. It can take on in-between values. It is not discrete.

C The number of votes polled by a politician in an election can be counted and cannot take on in-between values, so it is discrete.

D The number of coins in the pockets of students in a class can be counted and cannot take on in-between values, so it is discrete.

E The shoe sizes of students in a class can be counted and cannot take on in-between values, so it is discrete.

$\therefore$ **B**

3 The probabilities in parts A, B, C, D and E are all valid values.

A: $\Sigma p(x) = 1$

B: $\Sigma p(x) = 1\frac{1}{24}$

C: $\Sigma p(x) = 1$

D: $\Sigma p(x) = 1$

E: $\Sigma p(x) = 1$

In a probability distribution $\Sigma p(x) = 1$.

$\therefore$ **B**

4 For a uniform probability distribution $\mathrm{P}(X = x) = \frac{1}{n}$

$= \frac{1}{6}$

$= P(X = 7)$

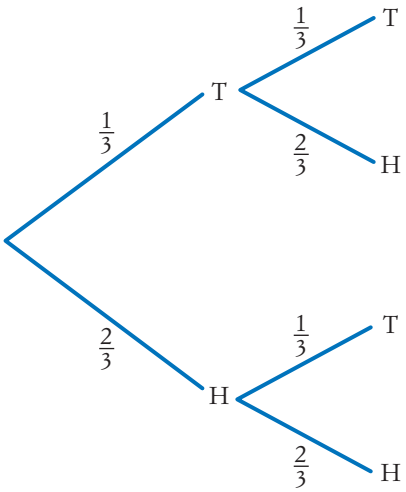
$\therefore$ **D**

5 $E(X) = 10 \times 0.15 + 20 \times 0.2 + 30 \times 0.25 + 40 \times 0.4$

$= 29$

$\therefore$ **C**

6 Tree diagram for this situation.



Let X = number of tails

$P(x = 0) = 1 \times \frac{2}{3} \times \frac{2}{3} = \frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$

$P(x = 1) = 2 \times \frac{2}{3} \times \frac{1}{3} = \frac{4}{9}$

$P(x = 2) = 1 \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$

$E(X) = 0 \times \frac{4}{9} + 1 \times \frac{4}{9} + 2 \times \frac{1}{9} = \frac{6}{9} = \frac{2}{3}$

$\therefore$ **C**

7 $E(X) = 2 \times 0.2 + 7 \times 0.25 + 11 \times 0.35 + 14 \times 0.2$

$= 8.8$

$E(X^2) = 2^2 \times 0.2 + 7^2 \times 0.25 + 11^2 \times 0.35 + 14^2 \times 0.2$

$= 94.6$

$\text{Var}(X) = 94.6 - 8.8^2 = 17.16$

$\therefore$ **C**

Part B – Short-answer questions

1 a Frequency table for the data.

x	1	2	3	4	5	6	7	8	9	10
f	1	3	2	4	7	4	4	5	3	1

[1 mark]

$\Sigma f = 1 + 3 + 2 + 4 + 7 + 4 + 4 + 5 + 3 + 1 = 34$ **[1 mark]**

x	1	2	3	4	5	6	7	8	9	10
f	1	3	2	4	7	4	4	5	3	1
$p(x)$	$\frac{1}{34}$	$\frac{3}{34}$	$\frac{2}{34}$	$\frac{4}{34}$	$\frac{7}{34}$	$\frac{4}{34}$	$\frac{4}{34}$	$\frac{5}{34}$	$\frac{3}{34}$	$\frac{1}{34}$

[1 mark]

b $4 \leq x < 6$ means $x = 4, 5$ or 6 **[1 mark]**

$P(4 \leq x < 6) = P(x = 4) + P(x = 5) + P(x = 6)$

$= \frac{4}{34} + \frac{7}{34} + \frac{4}{34}$

$= \frac{15}{34}$ **[1 mark]**

c $E(X) = \Sigma x \,.\, p(x)$ **[1 mark]**

$= 1 \times \frac{1}{34} + 2 \times \frac{3}{34} + 3 \times \frac{2}{34} + 4 \times \frac{4}{34} + 5 \times \frac{7}{34} + 6 \times \frac{4}{34}$

$+ 7 \times \frac{4}{34} + 8 \times \frac{5}{34} + 9 \times \frac{3}{34} + 10 \times \frac{1}{34}$

$= 5\frac{23}{34}$

≈ 5.68 **[1 mark]**

2 a Probability distribution.

x	1	4	6	7	8
$p(x)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

[1 mark]

b $E(X) = 1 \times \frac{1}{6} + 4 \times \frac{1}{3} + 6 \times \frac{1}{6} + 7 \times \frac{1}{6} + 8 \times \frac{1}{6}$

$= 5$ **[1 mark]**

c Let $Y = X + 5$

$E(Y) = E(X) + 5$ **[1 mark]**

$= 5 + 5$

$= 10$ **[1 mark]**

d Let $Z = 10X$

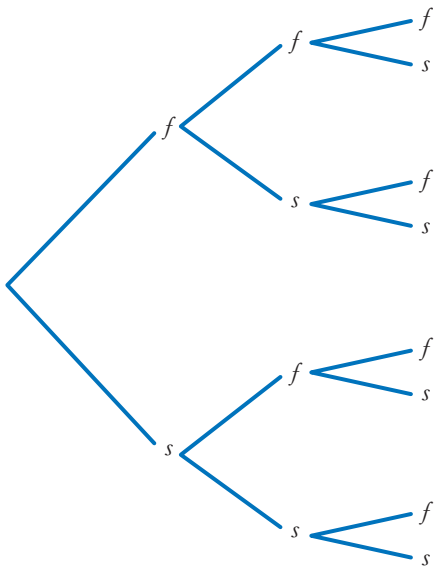
$E(Z) = 10E(X)$ **[1 mark]**

$= 10 \times 5$

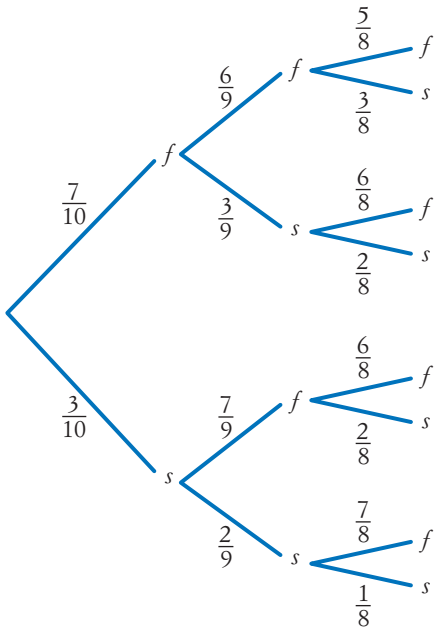
$= 50$ **[1 mark]**

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3 a Let defective chip = success (s) and non-defective chip = failure (f).



[1 mark]



[1 mark]

$P(\text{only 1 success}) = P(s, f, f) + P(f, s, f) + P(f, f, s)$ **[1 mark]**

$$= \frac{3}{10} \times \frac{7}{9} \times \frac{6}{8} + \frac{7}{10} \times \frac{3}{9} \times \frac{6}{8} + \frac{7}{10} \times \frac{6}{9} \times \frac{3}{8}$$

$= 0.525$ **[1 mark]**

b $P(\text{only success in 2nd selection}) = P(f, s, f)$ **[1 mark]**

$$= \frac{7}{10} \times \frac{3}{9} \times \frac{6}{8}$$

$= 0.175$ **[1 mark]**

4 a

x	3	4	5	6	7	8	9
$p(x)$	$\frac{2}{20}$	$\frac{2}{20}$	$\frac{4}{20}$	$\frac{4}{20}$	$\frac{4}{20}$	$\frac{2}{20}$	$\frac{2}{20}$

[2 marks]

b $E(X) = \Sigma x \,.\, p(x)$ **[1 mark]**

$= 3 \times \frac{2}{20} + 4 \times \frac{2}{20} + 5 \times \frac{4}{20} + 6 \times \frac{4}{20} + 7 \times \frac{4}{20} + 8 \times \frac{2}{20} + 9 \times \frac{2}{20}$ **[1 mark]**

$E(X) = 6$ **[1 mark]**

5 a Each of the outcomes is equally likely so the distribution is uniform. **[1 mark]**

b $n = b - a + 1$

$= 50 - 11 + 1$

$= 40$ **[1 mark]**

$P(X = x) = \frac{1}{n} = \frac{1}{40}$

$P(X = 17) = \frac{1}{40}$ **[1 mark]**

c $P(c \leq x \leq d) = \frac{d - c + 1}{b - a + 1}$ **[1 mark]**

$= \frac{32 - 21 + 1}{40}$

$= \frac{12}{40}$

$= \frac{3}{10}$ **[1 mark]**

6 $E(X) = \Sigma x \,.\, p(x)$ **[1 mark]**

$= 7 \times 0.09 + 9 \times 0.18 + 10 \times 0.33 + 12 \times 0.21 + 15 \times 0.19$

$= 10.92$ **[1 mark]**

$E(X^2) = 7^2 \times 0.09 + 9^2 \times 0.18 + 10^2 \times 0.33 + 12^2 \times 0.21 + 15^2 \times 0.19$

$= 124.98$ **[1 mark]**

$\text{Var}(X) = E(X^2) - E(X)^2$ **[1 mark]**

$= 124.98 - 10.92^2$

≈ 5.73 **[1 mark]**

$\sigma = \sqrt{\text{Var}(X)}$

$= \sqrt{5.7336}$

≈ 2.39 **[1 mark]**

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7 a Expected winnings $= E(X)$

$$E(X) = 1 \times \frac{24\,000}{240\,000} + 2 \times \frac{15\,000}{240\,000} + 3 \times \frac{12\,000}{240\,000} + 10 \times \frac{1200}{240\,000} + 20 \times \frac{500}{240\,000}$$

$$+ 30 \times \frac{300}{240\,000} + 500 \times \frac{4}{240\,000} + 5000 \times \frac{1}{240\,000}$$ **[1 mark]**

$$= 0.5333\ldots$$

Expected winnings $\approx$ 53 cents **[1 mark]**

b Expected value = Ticket price − expected winnings

$\approx$ 53 cents − \$1

$= -47$ cents **[1 mark]**

8 a Possible outcomes are:

0 heads = TTTT

1 head = HTTT, THTT, TTHT, TTTH

2 heads = HHTT, HTHT, HTTH, THHT, THTH, TTHH

3 heads = HHHT, HHTH, HTHH, THHH

4 heads = HHHH **[1 mark]**

$P(X = 0) = \frac{1}{16}$

$P(X = 1) = \frac{4}{16}$

$P(X = 2) = \frac{6}{16}$

$P(X = 3) = \frac{4}{16}$

$P(X = 4) = \frac{1}{16}$ **[1 mark]**

x	0	1	2	3	4
$P(X = x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

[1 mark]

b $E(X) = \Sigma x \,.\, p(x)$ **[1 mark]**

$$= 0 \times \frac{1}{16} + 1 \times \frac{4}{16} + 2 \times \frac{6}{16} + 3 \times \frac{4}{16} + 4 \times \frac{1}{16}$$

$= 2$ **[1 mark]**

The value of $E(X)$ agrees with your intuition, that if an unbiased coin is tossed repeatedly, about half of them should be heads. **[1 mark]**

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Part C – Analysis questions

1 Outcomes = HH, HT, TH, TT **[1 mark]**

Let X = the number of heads tossed.

x	0	1	2
$p(x)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

[1 mark]

Let W = winnings.

w	a	1	3
$p(w)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

[1 mark]

$E(W) = a \times \frac{1}{4} + 1 \times \frac{1}{2} + 3 \times \frac{1}{4}$ **[1 mark]**

For the game to be fair, $E(W) = 0$.

$\frac{a}{4} + \frac{1}{2} + \frac{3}{4} = 0$ **[1 mark]**

$\frac{a}{4} = -\frac{5}{4}$

$a = -5$ **[1 mark]**

For the game to be fair, the player should lose \$5 if no heads are tossed. **[1 mark]**

2 Required to prove: $\text{Var}(kX) = k^2\text{Var}(X)$ **[1 mark]**

$\text{Var}(X) = E(X^2) - [E(X)]$ **[1 mark]**

$\text{Var}(kX) = E((kX)^2) - [E(kX)]$ **[1 mark]**

$= E(k^2X^2) - [kE(X)]$ **[1 mark]**

$= k^2E(X^2) - k^2[E(X)]^2$ **[1 mark]**

$= k^2\{E(X^2) - [E(X)]^2\}$ **[1 mark]**

$= k^2\text{Var}(X)$

Q.E.D. **[1 mark]**

3 $E(X) = 0 \times 0.2 + 1 \times 0.3 + 2 \times 0.4 + 3 \times 0.1$

$= 1.4$ **[1 mark]**

Let annual cost = C

$E(C) = 2000 + 500 \times E(X)$ **[1 mark]**

$= 2000 + 500 \times 1.4$

$= 2700$ **[1 mark]**

The expected annual cost of servicing and repairing the multifunction device is \$2700. **[1 mark]**

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